

Computational modeling of neural populations

Part I: Neural fields with no spatial extension—Localized Wilson-Cowan model

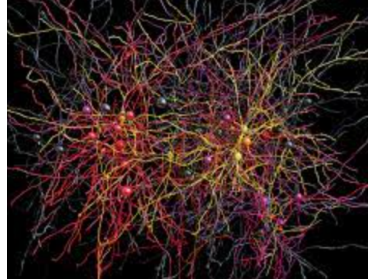
Biological motivation for neural population models

Sensory information is introduced to the nervous system as large-scale spatiotemporal activity in layers of neurons. Thus, the number of neurons involved is too high, it is not tractable for an approach starting from the single-cell level.

Many neural tasks are based on global processes. Approaches which emphasize only local (micro) properties are unlikely to give much insight.

It is a reasonable hypothesis that local interactions between neurons are mostly random, however, this randomness induces precise long-range interactions on average.

Modeling neural populations



Plouin et al.
H. of Cl. Neuro. 2013

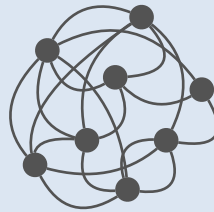
Networks

$$\frac{du_1}{dt} = f_1(u_1, \dots, u_n, t)$$

$$\frac{du_2}{dt} = f_2(u_2, \dots, u_n, t)$$

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$$\frac{du_n}{dt} = f_n(u_1, \dots, u_n, t)$$



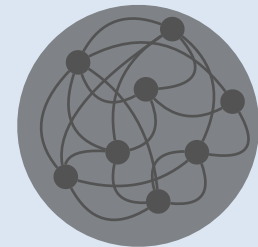
+ biophysically detailed
+ precise dynamics

- high dimensional
- difficult to manipulate
- computationally heavy

Coarse-grained approx.

$$\frac{dU}{dt} = F(U, t)$$

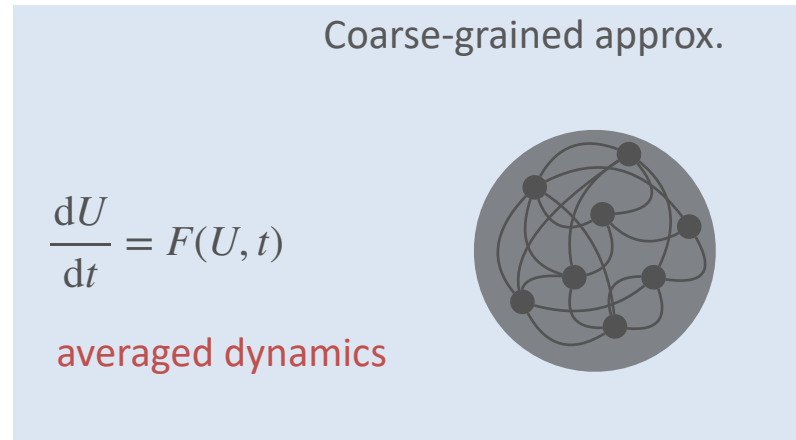
averaged dynamics



+ low dimensional
+ simple
+ computationally light

- not detailed
- approximate dynamics

Modeling neural populations



Mean field models (MFs)

Localized mean field: $\frac{dU}{dt} = F(U, t), \quad U(t)$

- Neurons in close proximity
- No spatial dependence
- Sometimes referred to as *neural mass* models

Spatially extended mean field: $\frac{\partial U}{\partial t} = F(U, x, t), \quad U(x, t)$

- Spatially distributed neurons
- Spatial dependence

Neural field models $\subset$ MFs

Localized neural fields:

- Wilson-Cowan '72

Spatially extended neural fields:

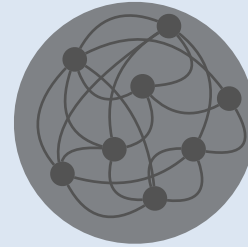
- Wilson-Cowan '73
- Amari '77

Modeling neural populations

Coarse-grained approx.

$$\frac{dU}{dt} = F(U, t)$$

averaged dynamics

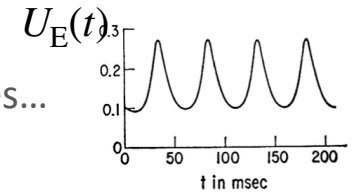


Neural field models

Localized neural fields:

- Wilson-Cowan '72

for temporal phenomena: limit cycles...



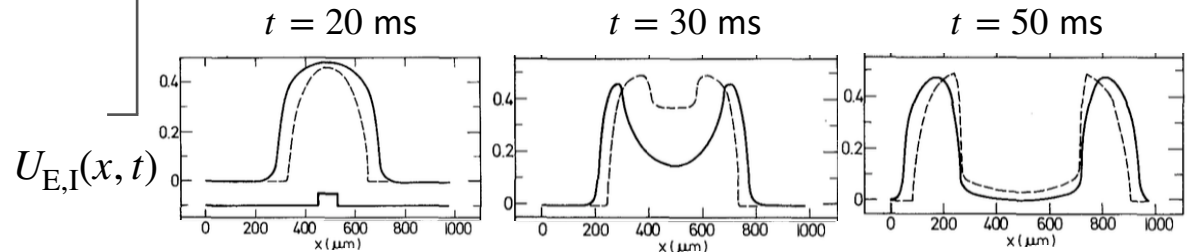
Wilson-Cowan 1972
Biophysical J

Spatially extended neural fields:

- Wilson-Cowan '73

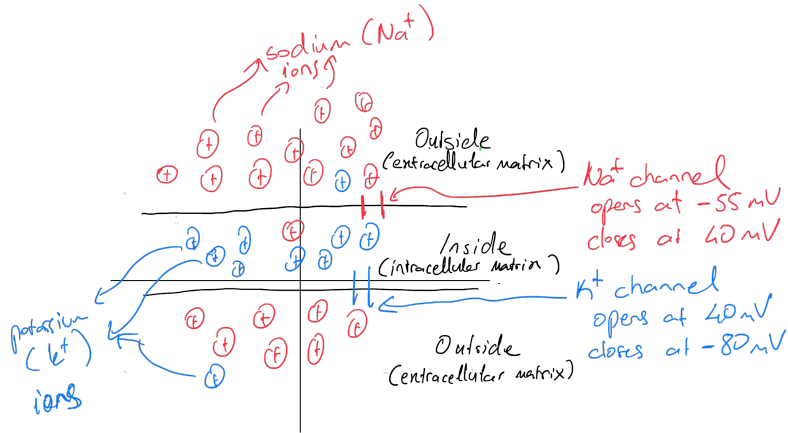
- Amari '77

for (also) spatio-temporal phenomena: traveling waves...



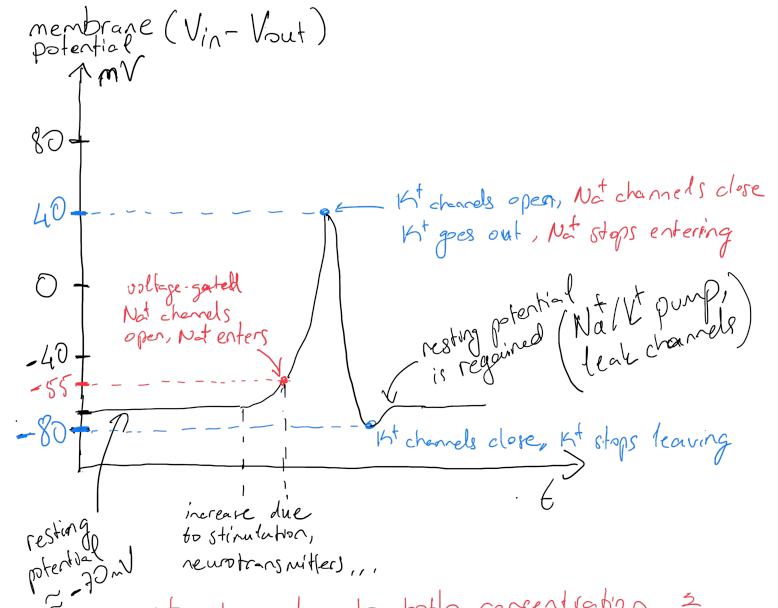
Wilson-Cowan 1973
Kybernetik

Intermezzo: action potential and synaptic interaction

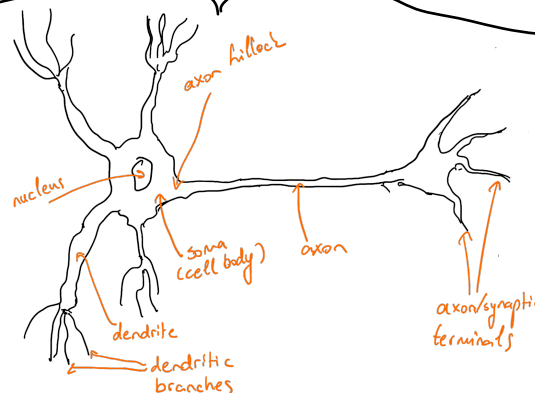


- * Na^+ higher concentration outside
lower " inside
- * K^+ higher concentration inside
lower " outside

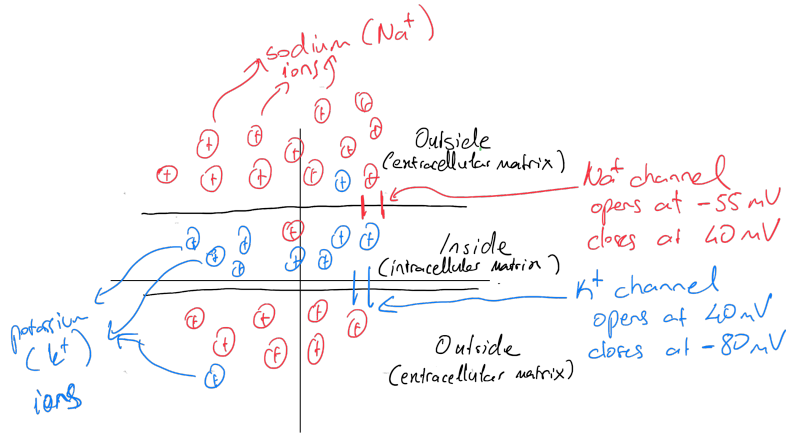
This equilibrium is maintained by the sodium-potassium pump (Na^+/K^+ -ATPase): Pumps 3 Na^+ out
" 2 K^+ in
Active $\rightarrow$ uses ATP



* Na^+ enters due to both concentration & electrical gradients
 K^+ leaves " " " "
 (it is repelled at $+40 \text{ mV}$)

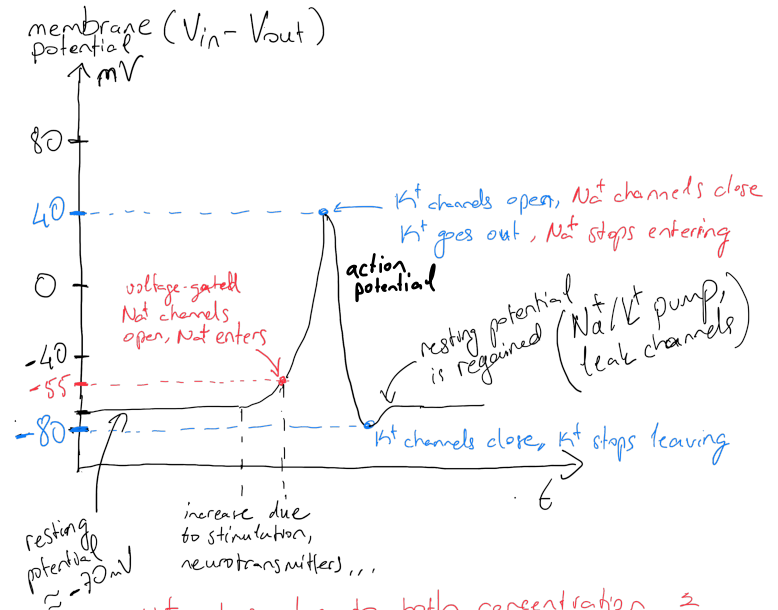


Intermezzo: action potential and synaptic interaction

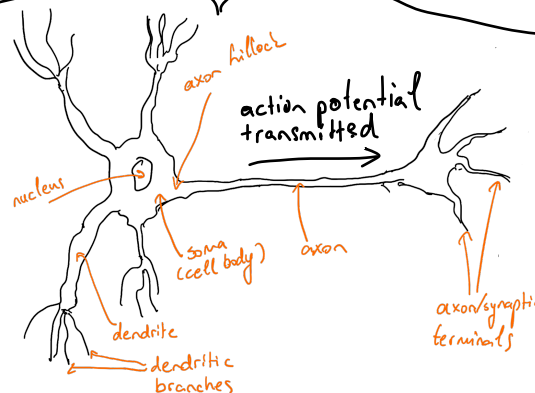


- * Na^+ higher concentration outside
lower " inside
- * K^+ higher concentration inside
lower " outside

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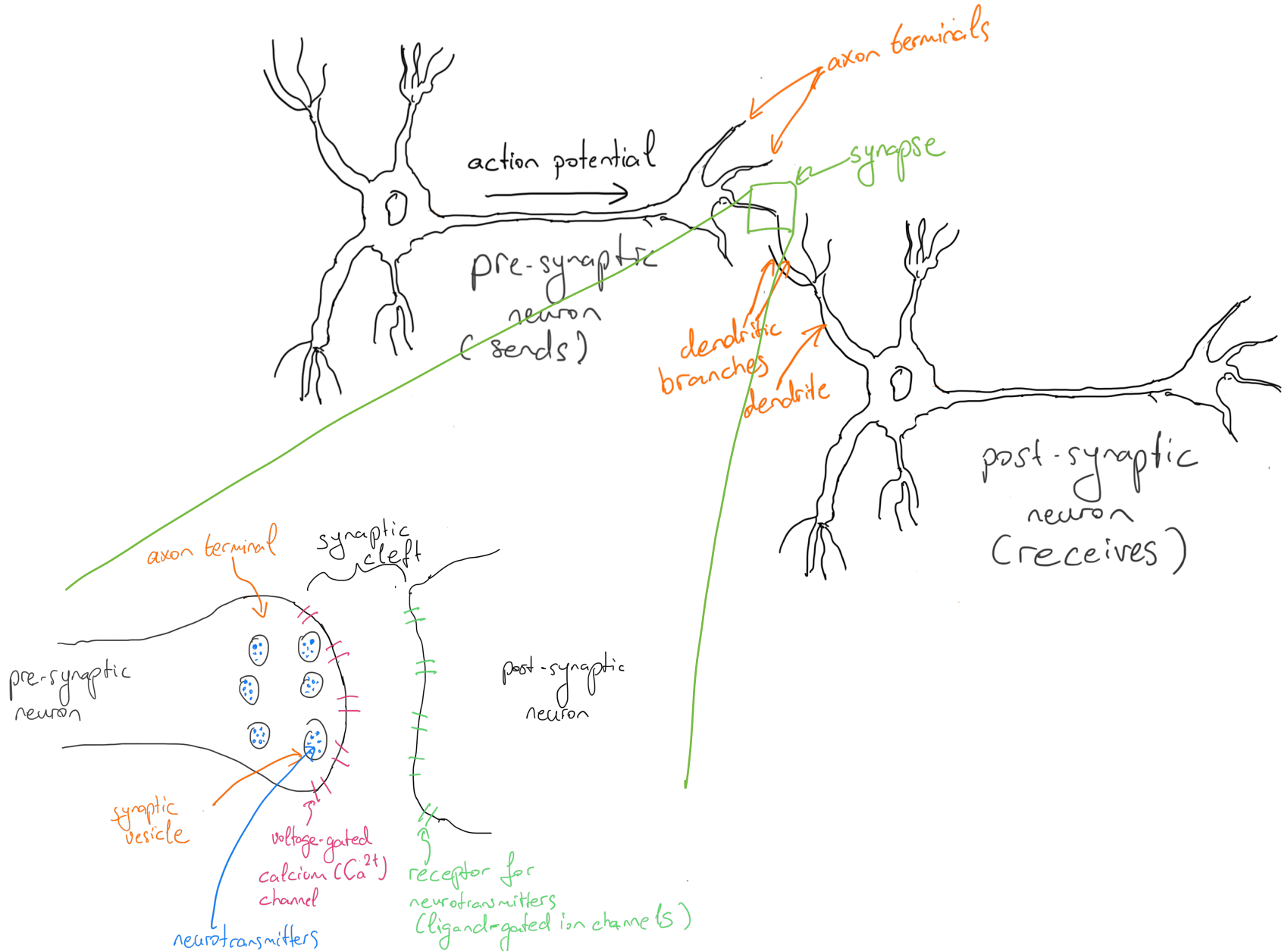


* Na^+ enters due to both concentration & electrical gradients
K⁺ leaves " " " "
(it is repelled at 40 mV)

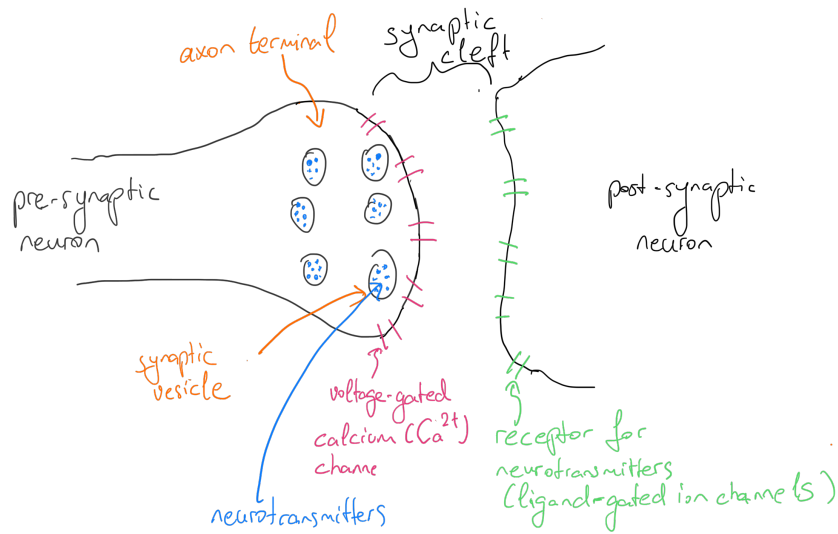


⚡⚡⚡ A short time after the action potential, the neuron CANNOT generate another action potential, due to the inactivation of voltage-gated Na⁺ channels. This time interval is called 'refractory period'.

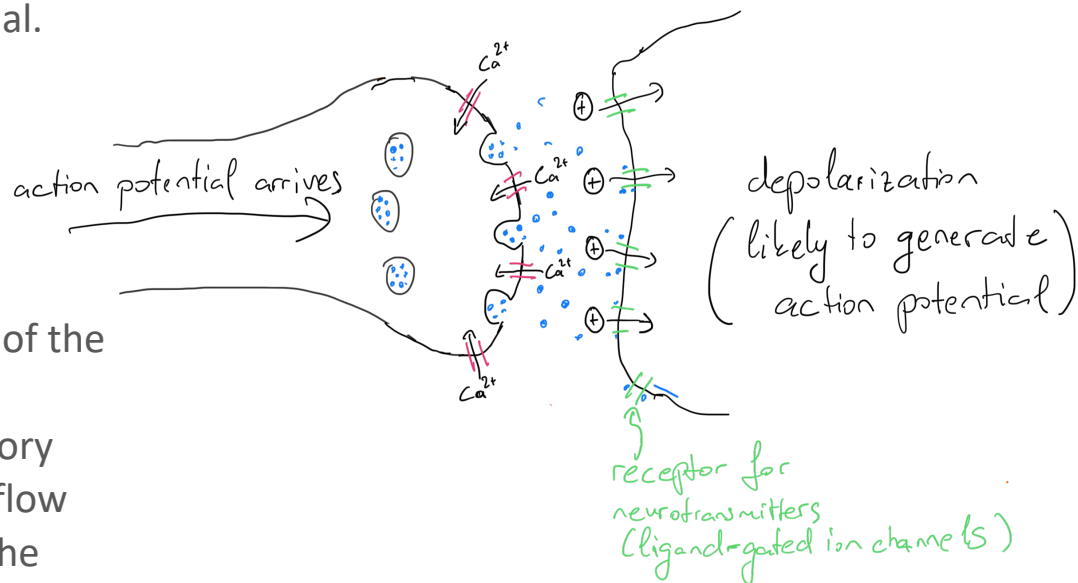
Intermezzo: action potential and synaptic interaction



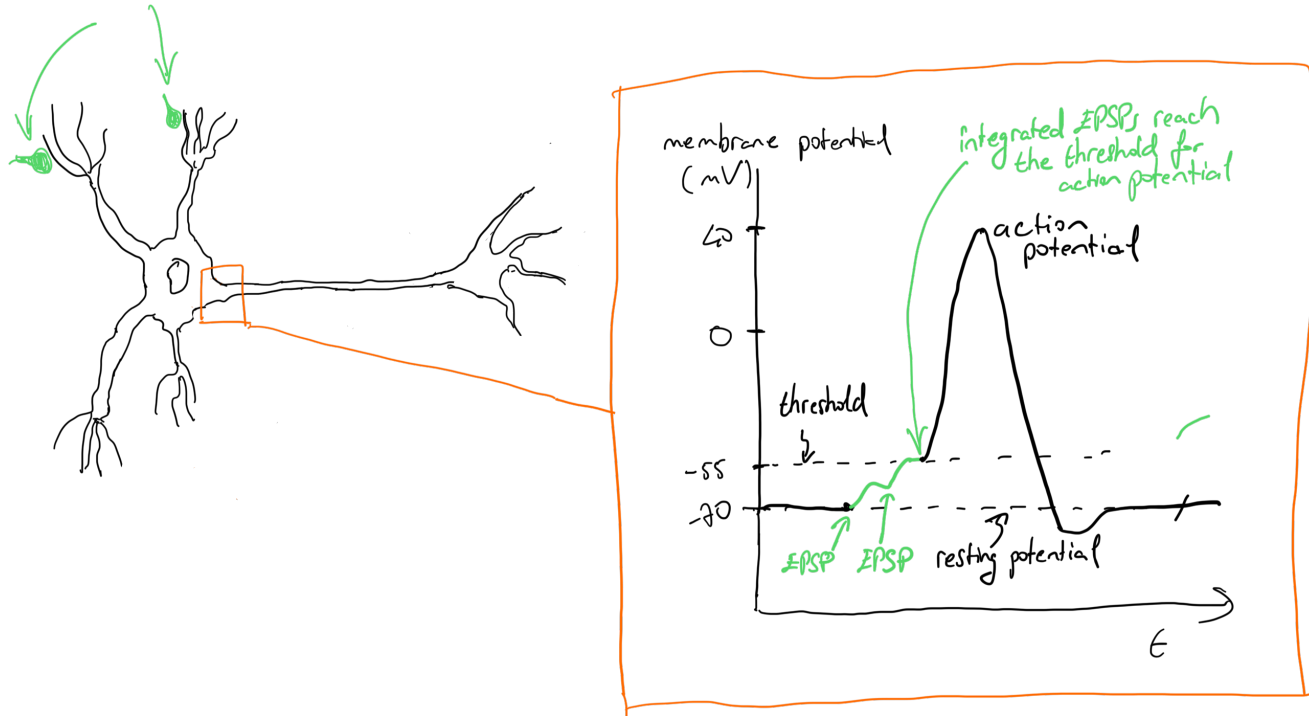
Intermezzo: action potential and synaptic interaction



- Action potential arrives to the axon terminal.
- It depolarizes the membrane.
- Voltage-gated Ca^{2+} channels open.
- Ca^{2+} enters in.
- This triggers the synaptic vesicles.
- Thus, they release neurotransmitters.
- The neurotransmitters attach to receptors of the target neuron.
- This causes positive ions to flow in (excitatory synaptic interaction) the target neuron or flow out (inhibitory synaptic interaction) from the target neuron.



Intermezzo: action potential and synaptic interaction



- Neurotransmitter causes ion channels open/close on the receiving neuron.
- This can result in a localized change in the membrane potential.
- If this change is depolarizing → excitatory post-synaptic potential (**EPSP**):
 - receiving neuron becomes more likely to fire an action potential
- If not → inhibitory post-synaptic potential (**IPSP**):
 - opposite of EPSP, receiving neuron becomes less likely to fire an action potential
- EPSP and IPSP interact through post-synaptic integration.
- This provides the key point in modeling the averaged synaptic interactions in neural fields.

Localized Wilson-Cowan model

Wilson-Cowan, Biophysical J, 1972

Modeling assumptions

1. Neurons are considered in close proximity and randomly connected. Yet, the connections are dense enough such that it is highly likely that there exists at least one path connecting any two neurons within the population. This is why the model is *localized*.
 - This allows us to neglect the spatial interactions, and to consider only the temporal dynamics of the population—> only time variable (t), **no** space variable (x).
2. Neural activity can be represented in terms of proportion of activated cells.
 - This implies that the related single neuron activity is not represented in terms of a single spike (action potential) but rather in terms of a normalized measure.
3. Neural processes are dependent on the interaction of excitatory and inhibitory neurons.
 - It is essential to take into account both excitatory and inhibitory neurons in any local population.

Localized Wilson-Cowan model

Wilson-Cowan, Biophysical J, 1972

Where do these equations come from?

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e \left(w_{ee} \bar{E} - w_{ei} \bar{I} + \bar{P} \right)$$

$$\mu \frac{d\bar{I}}{dt} = -\bar{I} + (1 - \bar{I}) S_i \left(w_{ie} \bar{E} - w_{ii} \bar{I} + \bar{Q} \right)$$

T : positive constant final time

$\bar{E}, \bar{I} : [0, T] \rightarrow [0, 1]$ proportion of firing excitatory ($\bar{E}$) or inhibitory ($\bar{I}$) neurons at the instant t

$\bar{P}, \bar{Q} : [0, T] \rightarrow \mathbb{R}$ external stimuli

S_e, S_i : nonlinear functions population transfer functions

$w_{ee}, w_{ei}, w_{ie}, w_{ii}$: positive constants connectivity weights

μ : positive constant time constant

Localized Wilson-Cowan model

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

Localized Wilson-Cowan model

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

Attention! $E(t)$ and $\bar{E}(t)$ are not the same thing! $I(t)$ and $\bar{I}(t)$ are not the same thing!

Localized Wilson-Cowan model

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

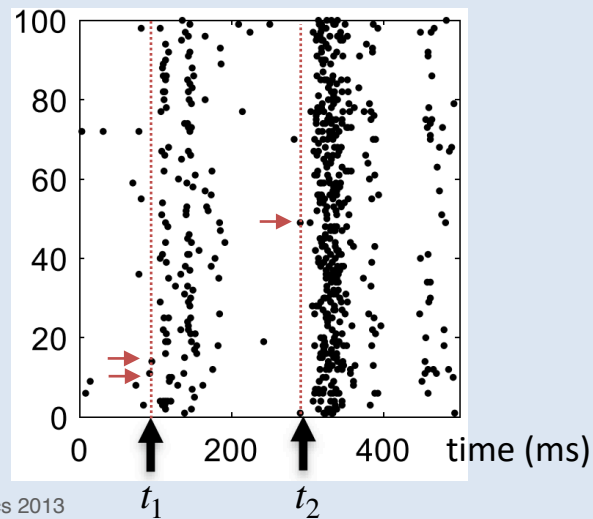
Localized Wilson-Cowan model

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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Example:

neuron index (excitatory)



Mounier et al.
Brain informatics 2013

$$E(t_1) = 0.02$$

$$E(t_2) = 0.01$$

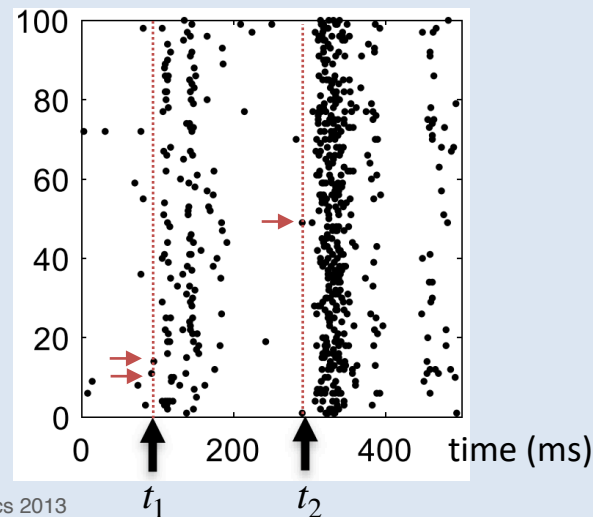
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Example:

neuron index (excitatory)



Mounier et al.
Brain informatics 2013

$$E(t_1) = 0.02$$

$$E(t_2) = 0.01$$

- $E(t) = I(t) = 0$ denotes the resting state (low-level background neural activity).
 - Resting state appears when there is no synaptic or external stimulation.
- In some variants of the model, negative values of $E(t)$ or $I(t)$ are attained.
 - This represents depression of resting activity.
- $E(t)$ and $I(t)$ are referred to as excitatory and inhibitory neural activity, respectively.
 - Referred to as such, it should be tacitly understood that they correspond to the proportions of firing neurons. **This might be different in other models!**

Localized Wilson-Cowan model

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- Let us assume that the refractory period has a duration of r ms.
- Then, the proportion of excitatory neurons which are refractory will be given by:

$$\int_{t-r}^t E(t') dt' .$$

- This means that the proportion of excitatory neurons which are sensitive is

$$1 - \int_{t-r}^t E(t') dt' .$$

- It is the same for the inhibitory neurons once we replace $E(t')$ with $I(t')$.

Localized Wilson-Cowan model: transfer functions

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e(w_{ee} \bar{E} - w_{ei} \bar{I} + \bar{P})$$

$$\mu \frac{d\bar{I}}{dt} = -\bar{I} + (1 - \bar{I}) S_i(w_{ie} \bar{E} - w_{ii} \bar{I} + \bar{Q})$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- Transfer functions $S_e(\xi)$ and $S_i(\xi)$ give the proportion of neurons which gets activated (fires) with respect to the synaptic input ξ that the subpopulation receives, **if none of the neurons were initially in the refractory period**.
- The general form of these functions can be derived by assuming the distribution of individual neural thresholds within a subpopulation is characterized by the distribution function $D(\theta)$ (e.g., normal distribution).
- If it is assumed also that the neurons receive the same numbers of excitatory and inhibitory synapses.
 - Then, on the average all neurons will receive the same average synaptic input $\xi(t)$.
- The subpopulation transfer function $S_e(\xi)$ (the same for $S_i(\xi)$) takes the following form:

$$S_e(\xi) := \int_{-\infty}^{\xi} D(\theta) d\theta.$$

Localized Wilson-Cowan model: transfer functions

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$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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- The subpopulation transfer function $S_e(\xi)$ (the same for $S_i(\xi)$) takes the following form:

$$S_e(\xi) := \int_{-\infty}^{\xi} D(\theta) d\theta.$$

- By construction, $S_e(\xi)$ is monotonically increasing and $0 \leq S_e(\xi) \leq 1$ ($D(\theta)$ is a distribution!).
- A typical choice for $D(\theta)$ is a unimodal function, e. g., Gaussian function.
- If $D(\theta)$ is unimodal, then $S_e(\xi)$ is a sigmoidal function, which is characterized as any function
 - that is monotonically increasing on $(-\infty, \infty)$,
 - that converges to 0 and 1 as $\xi \rightarrow -\infty$ and $\xi \rightarrow \infty$, respectively.
 - that has only one inflection point.

The same for $S_i(\xi)$ too!

Localized Wilson-Cowan model: transfer functions

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e(w_{ee}\bar{E} - w_{ei}\bar{I} + \bar{P})$$

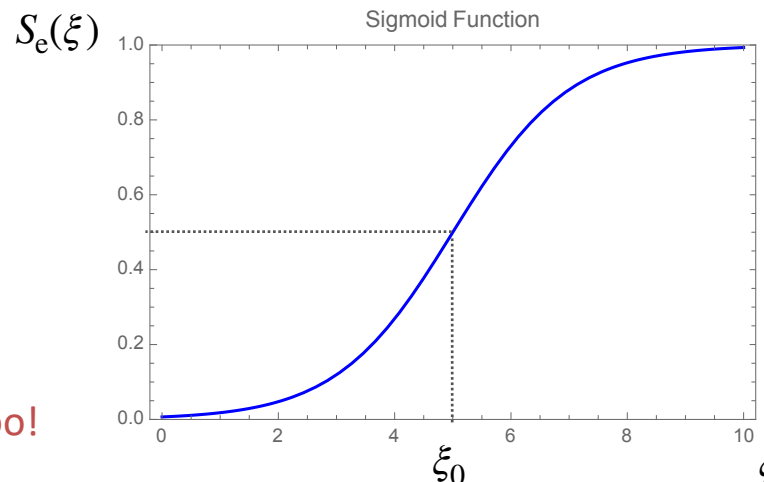
$$\mu \frac{d\bar{I}}{dt} = -\bar{I} + (1 - \bar{I}) S_i(w_{ie}\bar{E} - w_{ii}\bar{I} + \bar{Q})$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

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 - that converges to 0 and 1 as $\xi \rightarrow -\infty$ and $\xi \rightarrow \infty$, respectively,
 - that has only one inflection point.

Example:



$$S_e(\xi) = \frac{1}{1 + e^{-\beta(\xi - \xi_0)}}$$

$$\beta = 1, \quad \xi_0 = 5$$

β determines the sharpness of the nonlinearity: activity threshold heterogeneity of the population.

ξ_0 determines the population sensitivity to input.

The same for $S_i(\xi)$ too!

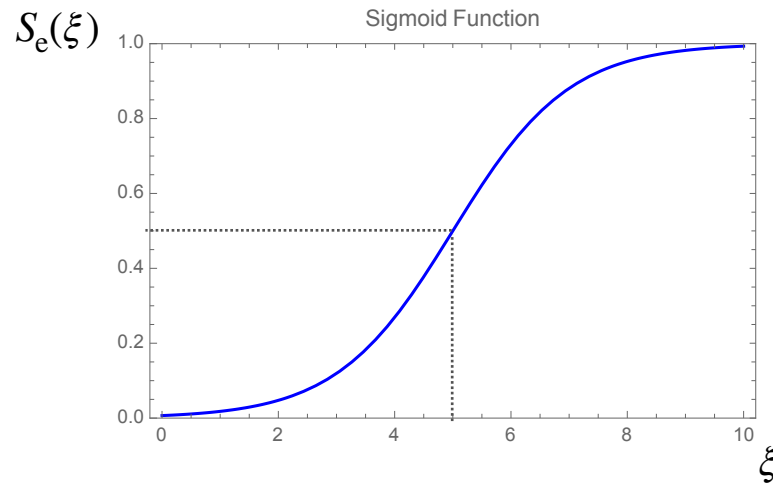
Localized Wilson-Cowan model: transfer functions

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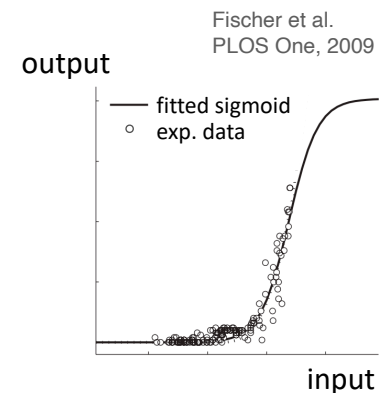
$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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Why sigmoid?

- Low excitation → no neurons activated: sigmoid models well.
- High excitation → all neurons activated: sigmoid models well.
- Experimental evidence: sigmoid-like input-output curves observed experimentally in neurons.



Localized Wilson-Cowan model: system equations

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e(w_{ee} \bar{E} - w_{ei} \bar{I} + \bar{P})$$

$$\mu \frac{d\bar{I}}{dt} = -\bar{I} + (1 - \bar{I}) S_i(w_{ie} \bar{E} - w_{ii} \bar{I} + \bar{Q})$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- Recall that individual neurons sum (integrate) their post-synaptic potentials (EPSPs + IPSPs) and the result constitutes the synaptic input. This input decays in time after it arrives to the target (receiving) neuron. We represent this decay with a function $\alpha(t)$. Then, the average level of synaptic input to an excitatory neuron at time t will be:

$$\xi(t) := \int_{-\infty}^t \alpha(t - t') (w_1 E(t') - w_2 I(t') + P(t')) dt'.$$

- The activity in a population at time $(t + \mu)$ is equal to the proportion of neurons which are both sensitive (not refractory, ready to fire) and which receive an average input above the activity threshold at time t :

$$E(t + \mu) = \left(1 - \int_{t-r}^t E(t') dt'\right) S_e(\xi(t)).$$

Similar for $I(t + \tau)$ too!

Localized Wilson-Cowan model: system equations

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e \left(w_{ee} \bar{E} - w_{ei} \bar{I} + \bar{P} \right)$$

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$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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- This provides the equations of the dynamics of localized neural populations as follows:

$$E(t + \mu) = \left(1 - \int_{t-r}^t E(t') dt' \right) S_e \left(\int_{-\infty}^t \alpha(t - t') (w_1 E(t') - w_2 I(t') + P(t')) dt' \right),$$

(Eq. 1)

$$I(t + \mu) = \left(1 - \int_{t-r}^t I(t') dt' \right) S_i \left(\int_{-\infty}^t \alpha(t - t') (w_3 E(t') - w_4 I(t') + Q(t')) dt' \right).$$

- This system is intuitive for physiological interpretation but complicated for mathematical treatment. Therefore, we simplify it via *time coarse graining*.

Localized Wilson-Cowan model: time coarse graining

$$\mu \frac{d\bar{E}}{dt} = -\bar{E} + (1 - \bar{E}) S_e(w_{ee} \bar{E} - w_{ei} \bar{I} + \bar{P})$$

$$\mu \frac{d\bar{I}}{dt} = -\bar{I} + (1 - \bar{I}) S_i(w_{ie} \bar{E} - w_{ii} \bar{I} + \bar{Q})$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- The idea is based on replacing the time-dependent variable, for example a function $f(t)$, with its moving time average over a suitably chosen interval s , $\bar{f}(t)$:

$$\bar{f}(t) := \frac{1}{s} \int_{t-s}^t f(t') dt'.$$

- Here s should be chosen suitably since this change of variable will average out rapid temporal variations happening in a shorter time scale than s .

Localized Wilson-Cowan model: time coarse graining

$$\begin{aligned} E(t + \mu) &= \left(1 - \int_{t-r}^t E(t') dt'\right) S_e \left(\int_{-\infty}^t \alpha(t-t') (w_1 E(t') - w_2 I(t') + P(t')) dt' \right) \\ I(t + \mu) &= \left(1 - \int_{t-r}^t I(t') dt'\right) S_i \left(\int_{-\infty}^t \alpha(t-t') (w_3 E(t') - w_4 I(t') + Q(t')) dt' \right) \end{aligned} \quad (\text{Eq. 1})$$

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- Here s should be chosen suitably since this change of variable will average out rapid temporal variations happening in a shorter time scale than s .
- Let us have a look how it applies to **(Eq. 1)**.

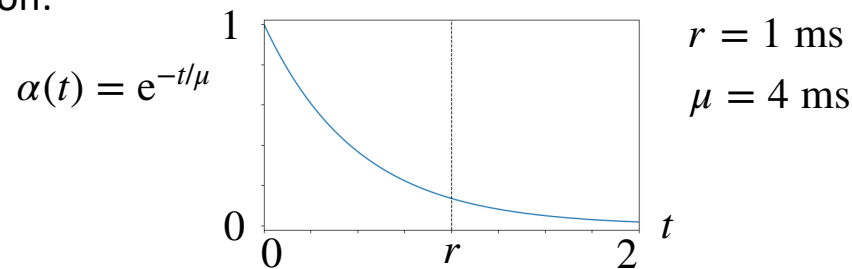
Localized Wilson-Cowan model: time coarse graining

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$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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- We can represent the time-decay $\alpha(t)$ of the synaptic input in terms of a function which is close to unity for $0 \leq t \leq r$ and vanishes quickly for $t > r$. A good choice can be an exponential decay function:



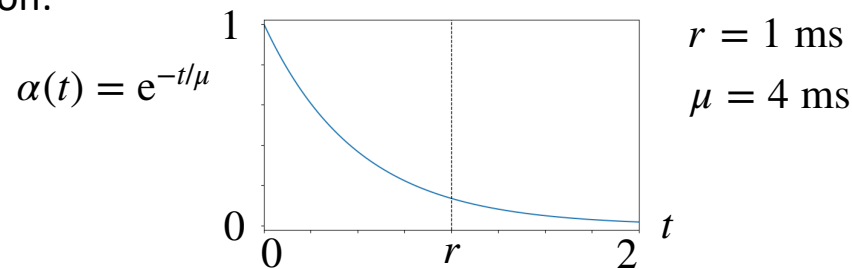
Localized Wilson-Cowan model: time coarse graining

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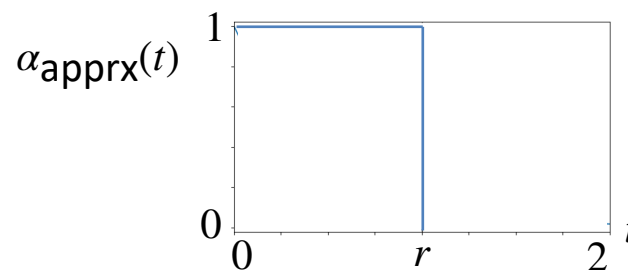
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- We can represent the time-decay $\alpha(t)$ of the synaptic input in terms of a function which is close to unity for $0 \leq t \leq r$ and vanishes quickly for $t > r$. A good choice can be an exponential decay function:



- This can be approximated by a step function:



Localized Wilson-Cowan model: time coarse graining

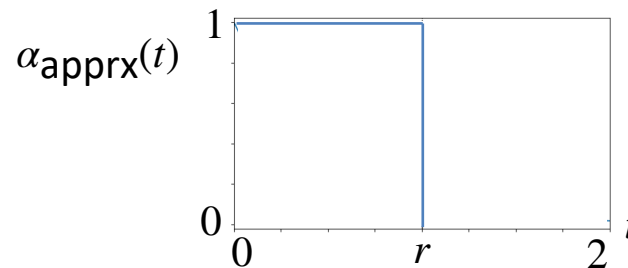
$$E(t + \mu) = \left(1 - \int_{t-r}^t E(t') dt'\right) S_e \left(\int_{-\infty}^t \alpha_{\text{apprx}}(t-t') (w_1 E(t') - w_2 I(t') + P(t')) dt' \right)$$

$$I(t + \mu) = \left(1 - \int_{t-r}^t I(t') dt'\right) S_i \left(\int_{-\infty}^t \alpha_{\text{apprx}}(t-t') (w_3 E(t') - w_4 I(t') + Q(t')) dt' \right)$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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- This can be approximated by a step function:



- With such choice for $\alpha(t)$, we can replace the time integrals on the right hand side with coarse-grained variables $\bar{E}(t)$ and $\bar{I}(t)$:

$$\int_{t-r}^t E(t') dt' \longrightarrow r \bar{E}(t),$$

$$\int_{-\infty}^t \alpha(t-t') E(t') dt' \approx \int_{t-r}^t E(t') dt' \longrightarrow r \bar{E}(t),$$

Similar for $I(t)$ too!

$$\int_{-\infty}^t \alpha(t-t') P(t') dt' \approx \int_{t-r}^t P(t') dt' \longrightarrow r \bar{P}(t).$$

Localized Wilson-Cowan model: time coarse graining

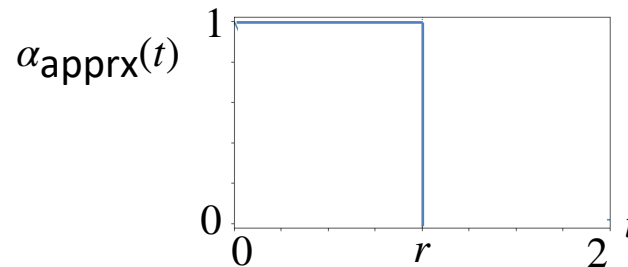
$$E(t + \mu) = (1 - r \bar{E}(t)) S_e(w_1 r \bar{E}(t) - w_2 r \bar{I}(t) + r \bar{P}(t))$$

$$I(t + \mu) = \left(1 - \int_{t-r}^t I(t') dt'\right) S_i\left(\int_{-\infty}^t \alpha(t-t') (w_3 E(t') - w_4 I(t') + Q(t')) dt'\right)$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- This can be approximated by a step function:



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Localized Wilson-Cowan model: time coarse graining

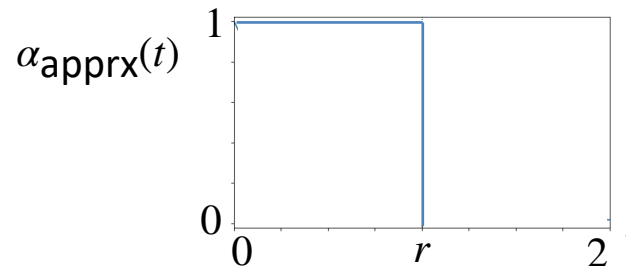
$$E(t + \mu) = (1 - r \bar{E}(t)) S_e(w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))$$

$$I(t + \mu) = \left(1 - \int_{t-r}^t I(t') dt'\right) S_i\left(\int_{-\infty}^t \alpha(t-t')(w_3 E(t') - w_4 I(t') + Q(t')) dt'\right)$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

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Similar for $I(t)$ too!

$$\int_{-\infty}^t \alpha(t-t') Q(t') dt' \approx \int_{t-r}^t Q(t') dt' \longrightarrow r \bar{Q}(t).$$

Localized Wilson-Cowan model: time coarse graining

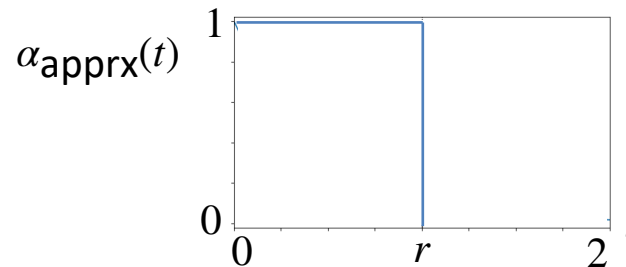
$$E(t + \mu) = (1 - r \bar{E}(t)) S_e(w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))$$

$$I(t + \mu) = (1 - r \bar{I}(t)) S_i(w_{ie} \bar{E}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t))$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- This can be approximated by a step function:



- With such choice for $\alpha(t)$, we can replace the time integrals on the right hand side with coarse-grained variables $\bar{E}(t)$ and $\bar{I}(t)$:

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Similar for $I(t)$ too!

$$\int_{-\infty}^t \alpha(t-t') Q(t') dt' \approx \int_{t-r}^t Q(t') dt' \longrightarrow r \bar{Q}(t).$$

Localized Wilson-Cowan model: time coarse graining

$$E(t + \mu) = (1 - r \bar{E}(t)) S_e(w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))$$

$$I(t + \mu) = (1 - r \bar{I}(t)) S_i(w_{ie} \bar{E}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t))$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- Finally, since time coarse graining has a smoothing effect on temporal variations, we should apply change of variables on the left hands side by replacing $E(t + \mu)$ and $I(t + \mu)$ with the Taylor expansions of the coarse-grained variable for small μ :

$$E(t + \mu) \approx \bar{E}(t) + \mu \frac{d\bar{E}(t)}{dt}$$

$$I(t + \mu) \approx \bar{I}(t) + \mu \frac{d\bar{I}(t)}{dt}$$

- This gives the final equations of the localized Wilson-Cowan model.

Localized Wilson-Cowan model: time coarse graining

$$\mu \frac{d\bar{E}(t)}{dt} = -\bar{E}(t) + (1 - r\bar{E}(t)) S_e(w_{ee}\bar{E}(t) - w_{ei}\bar{I}(t) + \bar{P}(t))$$
$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - r\bar{I}(t)) S_i(w_{ie}\bar{E}(t) - w_{ii}\bar{I}(t) + \bar{Q}(t))$$

$E(t)$: proportion of excitatory neurons firing per unit time at the instant t

$I(t)$: proportion of inhibitory neurons firing per unit time at the instant t

- Finally, since time coarse graining has a smoothing effect on temporal variations, we should apply change of variables on the left hands side by replacing $E(t + \mu)$ and $I(t + \mu)$ with the Taylor expansions of the coarse-grained variable for small μ :

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$$I(t + \mu) \approx \bar{I}(t) + \mu \frac{d\bar{I}(t)}{dt}$$

- This gives the final equations of the localized Wilson-Cowan model.

Localized Wilson-Cowan model: time coarse graining

$$\mu \frac{d\bar{E}(t)}{dt} = -\bar{E}(t) + (1 - r\bar{E}(t)) S_e(w_{ee}\bar{E}(t) - w_{ei}\bar{I}(t) + \bar{P}(t))$$
$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - r\bar{I}(t)) S_i(w_{ie}\bar{E}(t) - w_{ii}\bar{I}(t) + \bar{Q}(t))$$

$\bar{E}(t)$: time-coarse representation of proportion of excitatory neurons firing at the instant t

$\bar{I}(t)$: time-coarse representation of proportion of inhibitory neurons firing at the instant t

- Finally, since time coarse graining has a smoothing effect on temporal variations, we should apply change of variables on the left hands side by replacing $E(t + \mu)$ and $I(t + \mu)$ with the Taylor expansions of the coarse-grained variable for small μ :

$$E(t + \mu) \approx \bar{E}(t) + \mu \frac{d\bar{E}(t)}{dt}$$

$$I(t + \mu) \approx \bar{I}(t) + \mu \frac{d\bar{I}(t)}{dt}$$

- This gives the final equations of the localized Wilson-Cowan model.

Localized Wilson-Cowan model: time coarse graining

$$\mu \frac{d\bar{E}(t)}{dt} = -\bar{E}(t) + (1 - \bar{E}(t)) S_e(w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))$$

$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - \bar{I}(t)) S_i(w_{ie} \bar{E}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t))$$

$\bar{E}(t)$: time-coarse representation of proportion of excitatory neurons firing at the instant t

$\bar{I}(t)$: time-coarse representation of proportion of inhibitory neurons firing at the instant t

- Finally, since time coarse graining has a smoothing effect on temporal variations, we should apply change of variables on the left hands side by replacing $E(t + \mu)$ and $I(t + \mu)$ with the Taylor expansions of the coarse-grained variable for small μ :

$$E(t + \mu) \approx \bar{E}(t) + \mu \frac{d\bar{E}(t)}{dt}$$

$$I(t + \mu) \approx \bar{I}(t) + \mu \frac{d\bar{I}(t)}{dt}$$

- This gives the final equations of the localized Wilson-Cowan model.
- In many cases, $r = 1$ can be used. This is also supported by experimental studies showing that r is around 1-2 ms [Eccles, The Physiology of Synapses, 1964].

Localized Wilson-Cowan model: description

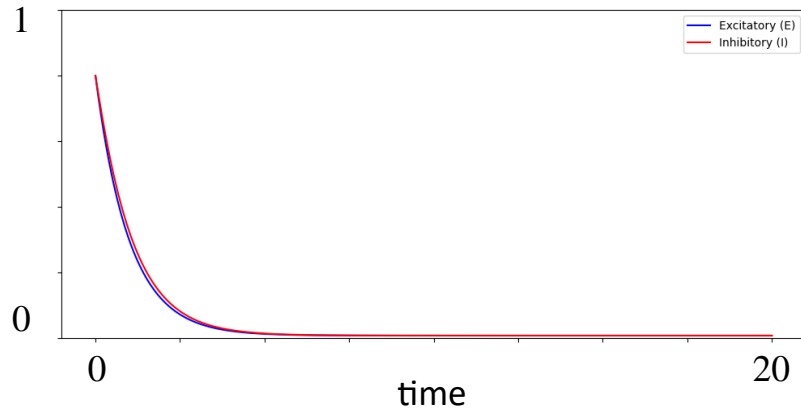
$$\mu \frac{d\bar{E}(t)}{dt} = \underbrace{-\bar{E}(t)}_{\text{negative feedback}} + \underbrace{(1 - \bar{E}(t)) S_e (w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))}_{\substack{\text{total input} = \text{synaptic interactions} \\ + \text{external stimulation}}}$$

$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - \bar{I}(t)) S_i (w_{ie} \bar{E}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t))$$

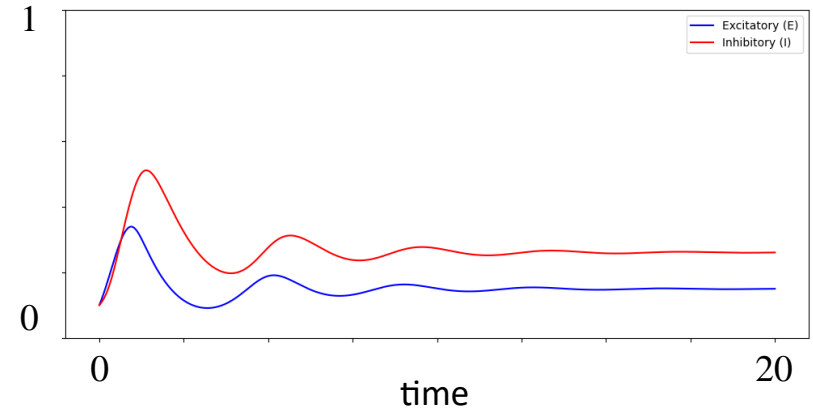
Negative feedback ensures that the population comes to the resting state and remains there in the absence of any synaptic and external input.

Localized Wilson-Cowan model: temporal activity patterns

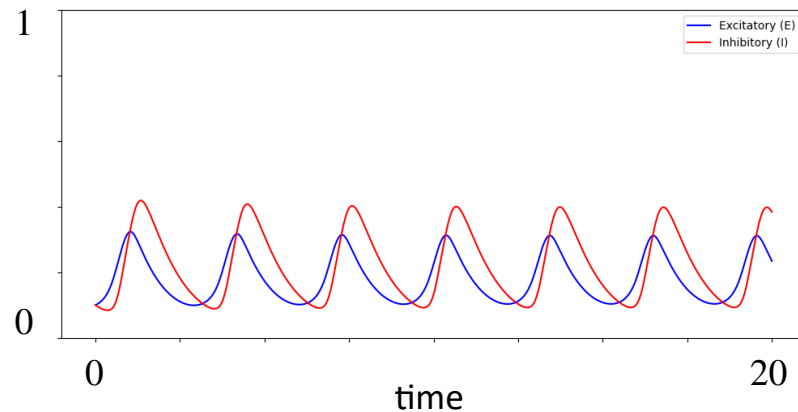
$\bar{E}(t), \bar{I}(t)$ Resting state



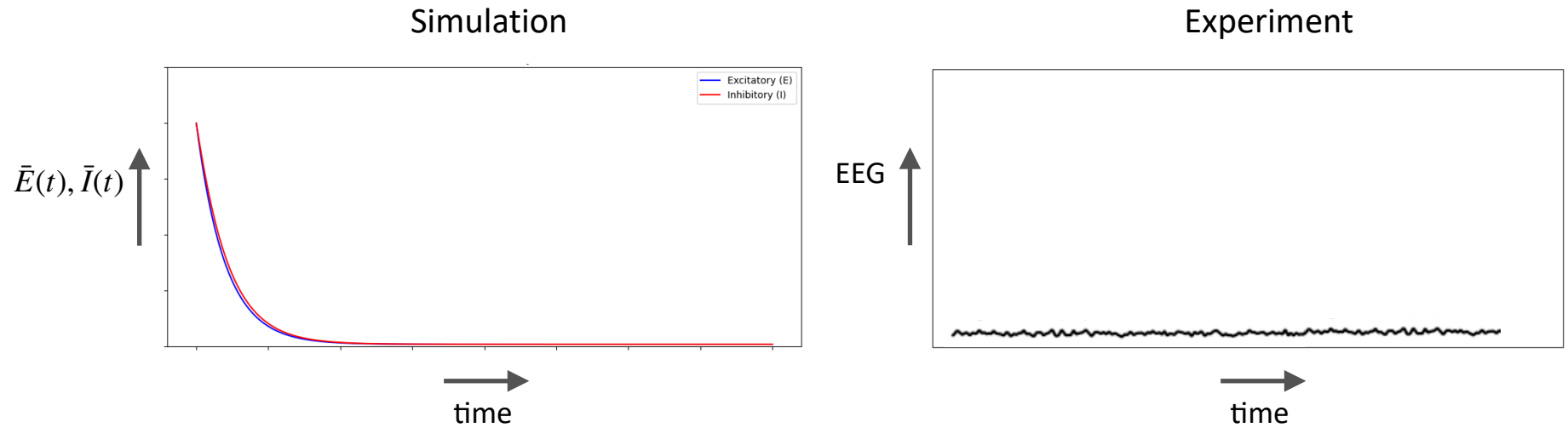
$\bar{E}(t), \bar{I}(t)$ Damped oscillations



$\bar{E}(t), \bar{I}(t)$ Limit cycle



Example: resting state

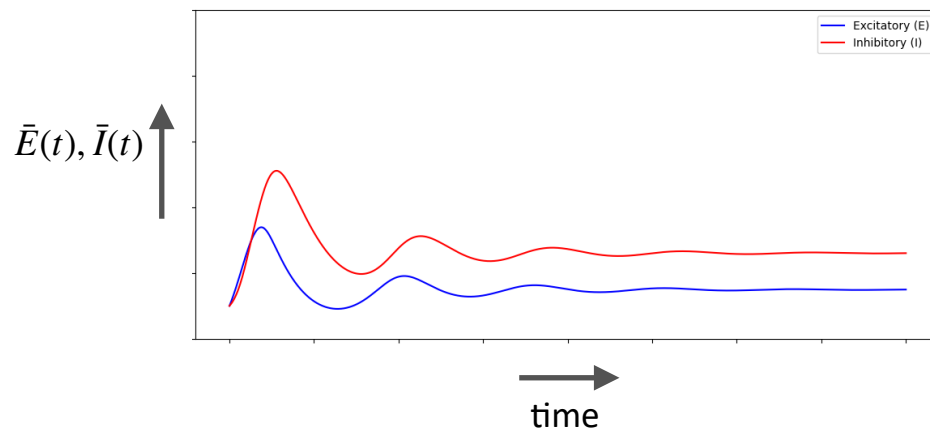


adapted from
<https://frederic-vw.github.io/eeg-patterns/>

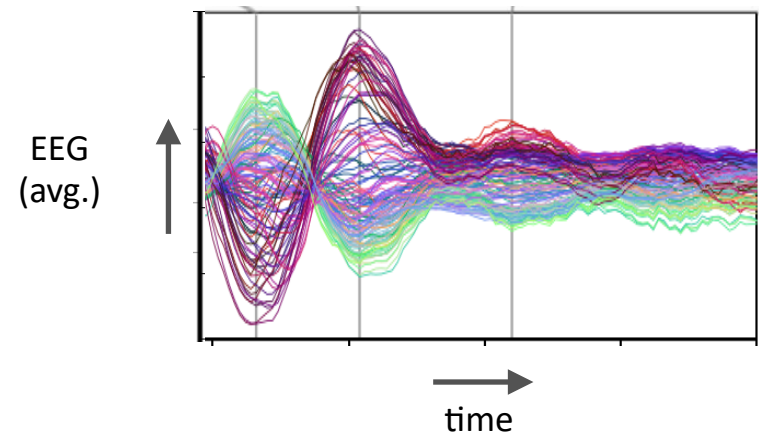
Commonly observed when there is no stimulus, e.g.,
in task-free experiments, where eyes are closed.

Example: damped oscillations

Simulation



Experiment (human)

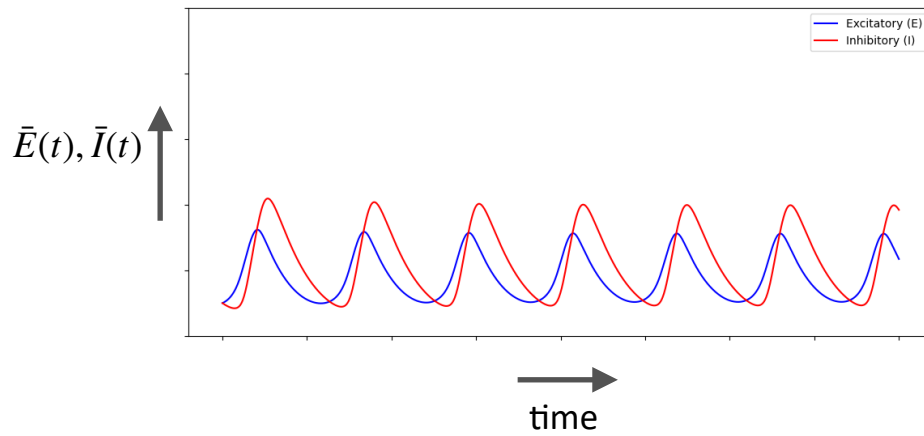


Pascucci et al., Sci Data, 2022

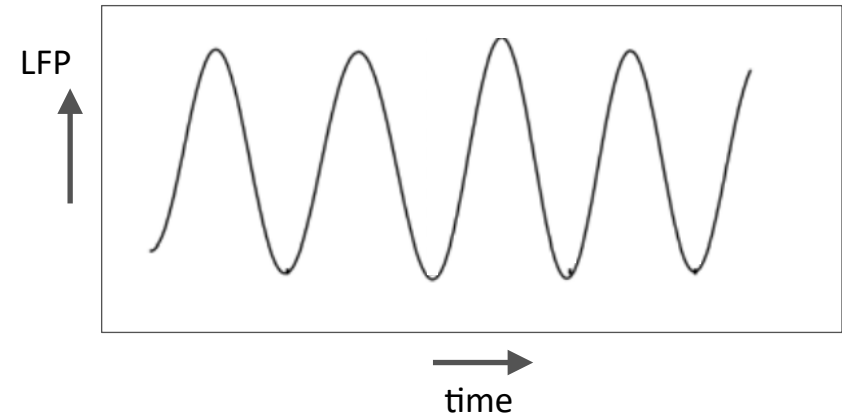
Commonly observed in stimulus-evoked activity

Example: limit cycle

Simulation



Experiment (mouse)



Scheffzük et al., PLOS One, 2011

Observed during several cognitive tasks, and in different brain states, e. g., in REM sleep.

$$\mu \frac{d\bar{E}(t)}{dt} = \underbrace{-\bar{E}(t)}_{\text{negative feedback}} + \underbrace{(1 - \bar{E}(t)) S_e(w_{ee} \bar{E}(t) - w_{ei} \bar{I}(t) + \bar{P}(t))}_{\text{synaptic interactions external stimulation}}$$

$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - \bar{I}(t)) S_i(w_{ie} \bar{E}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t))$$

- Localized Wilson-Cowan is one of the first neural field type models.
- It has no spatial dependence.
- It shows the excitatory-inhibitory interactions.
- It is used for temporal phenomena, not for spatial phenomena.
- It provides a starting point for spatially-extended Wilson-Cowan model.

The end